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The Role of Implicit and Explicit Beliefs in Grave-Good Practices: Evidence for Intuitive Afterlife Reasoning

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Abstract

The practice of burying objects with the dead is often claimed as some of the earliest evidence for religion, on the assumption that such "grave goods" were intended for the decedents' use in the afterlife. However, this assumption is largely speculative, as the underlying motivations for grave-good practices across time and place remain little understood. In the present work, we asked if explicit and implicit religious beliefs (particularly those concerning the continuity of personal consciousness after death) motivate contemporary grave-good practices. Across three studies, and comparing participants from the United States and NZ, we measured grave-good deposition at actual or hypothetical funerals, finding that jewelry, photographs, and other items with sentimental, emotional, and relationship value were common. In addition, *intuitive afterlife reasoning* (as measured by people's attributions of mental states to the dead) motivated grave-good decision-making for about half (Study 2) or more (Study 3) people, including afterlife nonbelievers ("extinctivists"), while those who held explicit (i.e., stated) afterlife beliefs were more likely to participate in the practice. The decision to leave grave goods was also associated with magical contagion beliefs and a need for personal comfort, while other motivations, such as social signaling, were less common. Our results suggest that "afterlife use" is a common motivation for grave-good practices, and that humans possess evolutionarily early intuitions about postdeath consciousness.

Keywords: Grave goods; Burial objects; Religion; Afterlife belief; Intuitive belief; Simulation constraint; Magical contagion; Nonbelief

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1. Introduction

People have been burying objects with the dead for longer than recorded history. Excavations of the Qafzeh and Skhul Caves in Israel, for example, have uncovered graves as old as 100,000 years that contain flint artifacts, animal bones, seashells, and lumps of red ochre (Bar-Yosef Mayer, Vandermeersch, & Bar-Yosef, 2009). By 25,000 years ago, burials included decorative clothing as well as jewelry, weapons, animal carvings, and polished pebbles made of stone, ivory, or bone (Hovers & Belfer-Cohen, 2018).

Although there are many possible reasons for depositing objects and artifacts with the dead, evidence suggests that some "grave goods" were intended to be used in, or on the journey to, the afterlife (Kellehear, 2007; Paulsen, 1967). Ancient Egyptians, for example, buried their dead with maps and sacred texts to guide their passage; they also included food for sustenance and Shabti dolls (funerary figures) to do chores in the afterlife (Ikram, 2015). Similarly, Romans buried their dead with a coin for Charon, the ferryman to the underworld; Scandinavians buried their dead with drinking vessels and military equipment (see *Beowulf* and *Egil's Saga*); and the Chinese have a long tradition of burning paper money and models of precious objects for the deceased to take with them in the afterlife (Härke, 2014).

Today, despite differences in beliefs about what death entails for the individual, it is still common to bury the dead with contemporary items of sentimental value, such as jewelry, photographs, special items of clothing, consumables (e.g., cigars and alcohol), stuffed animals, religious items, flowers, and written messages (Elliot, 1990; Harper, 2012). However, explanations for *why* these items are deposited with the decedent are diverse. For example, one sample of American morticians reported that the primary motivation for grave goods was "for the sentiment of relatives," with "implications in the afterlife" being only secondary (Elliot, 1990). In another study, however, Harper (2012) found (based on direct observation of American and British funeral homes) that grave-good explanations sometimes implied afterlife beliefs, not unlike those of ancient populations. For example, one bereaved person said that they had included photographs so that their parents "felt safe" along their journey. Another individual, who left a cigar, commented that the deceased "would have to get a light off someone else up there" (Harper, 2012, p. 56).

On the surface, such historically and culturally recurring funerary practices seem to suggest that humans have a predisposition to believe in, and prepare for, an afterlife. Until recently, however, Christian burials included few or no grave goods, presenting a challenge to the "afterlife use" theory (i.e., that grave goods are of physical or emotional use to the deceased after death). This absence may reflect a rejection of earlier pagan practices (Härke, 2014), a preference for abstinence (Baumard, Hyafil, Morris, & Boyer, 2015), or a belief that the afterlife will be either a paradise in which everything is provided or a hellish underworld in which nothing can be enjoyed. Furthermore, the apparent resurgence in grave-good practices in the Christianized world (Elliot, 1990; Harper, 2012) appears to be inconsistent with the growing number of *extinctivists* (those who believe there is no continuity of personal consciousness after death; see Bering, 2006): in the past 75 years, surveys suggest that afterlife belief has decreased from 76% to 66% in the United States, 49% to 35% in the UK, and 84% to 49% in Canada (Bibby, 2017). It may be that, contrary to the afterlife use theory,

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grave-good deposition is driven by other symbolic, emotional, and practical reasons that have little to do with a belief in immortal souls. Such practices may serve a functional purpose for the living, providing psychological comfort (Elliot, 1990), demonstrating wealth or status (Härke, 2014), or signaling alignment with group norms (Sosis & Alcorta, 2003). However, Harper (2012) observed that grave goods were sometimes left privately (e.g., letters in sealed envelopes), suggesting that social and reputational motives cannot completely explain the grave-good phenomenon.

Another possibility is that "magical contagion" beliefs motivate the deposition of grave goods that are purported to bear the deceased's essence (Nemeroff & Rozin, 1994). This can include the disposal of inalienable property that is too strongly associated with the deceased to be inherited or sold, such as clothing, jewelry, and spectacles (Härke, 2014; Harper, 2012). At burial, the bereaved may consider such objects an extension or inseparable part of the deceased's identity, such that if "all" of the deceased is to be buried in order to complete the funerary ritual, then any objects that are "imbued with their essence" (Klevnäs, 2016) or "polluted" by it (Härke, 2014) must be included, or else kept to retain some "remnant" of the deceased in the world of the living (e.g., celebrity memorabilia; Gibson, 2010; Newman, Diesendruck, & Bloom, 2011). Magical contagion may even explain grave goods that have not been physically touched by the deceased (Huang, Ackerman, & Newman, 2017), such as the cigar from the earlier example, or other objects that characterize the deceased's identity, personal preferences, physical features, or relationships (Drenten, McManus, & Labrecque, 2017; Layne, 2000), or that are symbolic representations of their beliefs or biography, such as a Christian crucifix, or soil from their homeland (Härke, 2014). In other words, grave goods may be necessary to fully reconstruct the essence of a person at burial, and, in modern times, this motivation may be especially prevalent, given consumerist attitudes about the expression and expansion of personhood through possessions (Belk, 1988; Newman & Smith, 2016).

Magical contagion could also explain why pets, which may not be thought of as going to an afterlife, are buried with their own grave goods (e.g., a favorite toy), and such practices could serve simultaneously to remove material triggers of grief. Consistent with this "reminder theory" (Walter, 1999, pp. 26–28), one of Elliot's (1990) respondents reported that grave goods should be "out of sight so as to not remind them of their loss" (p. 609), with the author concluding that grave goods are an attempt "to put [the] final touches on a relationship" (p. 611). Similarly, several North American tribes destroyed the house of the deceased and banned the mention of their name, while Bronze Age peoples appear to have hidden grave goods from view at funerals (Härke, 2014).

Thus, several reasons for grave-good practices appear in particular cultures at particular times, and several may be relevant to contemporary funerals. However, we propose that these reasons are either secondary or related to afterlife use, and that the growing population of extinctivists is, in fact, not at odds with this primary function when considering the ongoing popularity of grave-good practices. We propose that, despite extinctivists' self-reports, most people hold "implicit" beliefs about the afterlife; that is, they are intuitively disposed to foresee themselves, and their loved ones, as continuing to exist as mental agents after death. For example, the paucity of grave goods at Neanderthal burial sites has been interpreted as evidence of their lack of episodic memory—the capacity to envisage a time after death (Petru,

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2018). Conversely, it may be that human cognition makes it exceptionally difficult, if not impossible, to properly envision death as a psychological end (Freud, 1915; Nichols, 2007), what Bering (2002, 2006) refers to as a "simulation constraint" on humans' ability to mentally represent death as an absence of consciousness. Moreover, self-identified extinctivists may behave in a more cautious manner when future costs or rewards are significant, taking precious objects into whatever *might* lie beyond, and, therefore, betraying context-dependent cognitive biases or unarticulated thoughts about life after death.

There is some empirical support for the notion that humans reason implicitly that consciousness continues after death. In a study by Bering (2002), for instance, many individuals who identified as extinctivists nevertheless assumed that a dead person would retain their psychological capacities (e.g., that the person would "know" that they were dead). Even participants who consistently rejected the possibility of postmortem emotional and epistemic states struggled when reasoning about them (e.g., it took them a relatively long time to decide whether the deceased could believe or love). A similar study with children showed that reasoning in terms of psychological continuity after death *decreases* with age (presumably as a consequence of materialist-based science education), suggesting that afterlife beliefs are "cognitive defaults" rather than products of religious enculturation (Bering & Bjorklund, 2004). These findings have since been replicated across different cultural contexts (e.g., Bering, Hernandez-Blási, & Bjorklund, 2005a; Georgiadou & Pnevmatikos, 2019; Misailidi & Kornilaki, 2015; but see Barrett et al., 2021; Watson-Jones, Busch, Harris, & Legare, 2017).

Other studies have also revealed a discrepancy between what people say they believe and how they behave or reason about the afterlife. For example, Haidt and colleagues (2000) observed that extinctivists were reluctant to sell their soul to the experimenter, even when told that the contract was not legally binding and would be shredded later (see also Preston, Ritter, & Hepler, 2013). In another study, Bering and colleagues (2005b) found that individuals who were casually informed that a ghost had been spotted in the laboratory were less likely than control participants to cheat on a competitive task when left alone (although afterlife beliefs were not measured). In a recent follow-up, Bering, Smith, Stojanov, Hughes, and Halberstadt (2022) examined people's responses to being alone in a room in which it was claimed (or, in a control condition, not mentioned) that a university employee had recently died and that his ghost had been spotted there. Participants in the "ghost" condition were more distressed (according to measures of skin conductance), and reported being more afraid, when a light in the room went off for several seconds. Notably, this effect was independent of their explicit afterlife beliefs.

Thus, at the core of the present work is the question of whether grave goods are motivated primarily by afterlife reasoning (the "afterlife use" theory), even among those who explicitly reject the existence of an afterlife. In the following studies, we contrast participants' explicit religiosity or stated afterlife beliefs with their grave-good preferences and motivations at actual (New Zealand) or hypothetical (the United States) funerals. Our cultural focus provides an appropriate setting for our investigation of implicit afterlife beliefs (i.e., intuitive afterlife reasoning), drawing on the relatively high frequency of extinctivists/skeptics in the Western world. Our first study sought to establish the frequency of grave-good deposits, the types of objects buried (or burned, in the case of cremation) with the dead, and who is participating

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in this practice in New Zealand, which ranks among the "least religious" countries worldwide (just after the Netherlands; Gallup, 2008). Two further experimental studies served to tease apart people's underlying reasons for wanting to be personally buried with grave goods (Study 2) or to leave grave goods for others (Study 3). We predicted that, although explicit religious and/or afterlife beliefs would be associated with a greater stated desire for grave goods, extinctivists and the nonreligious would nevertheless still show evidence of intuitive afterlife reasoning.

2. Study 1

2.1. Method

The following study (and subsequent studies) received ethical approval from the University of Otago Human Ethics Committee (DP 19/18) and adhered to the principles of the Declaration of Helsinki. Contacts from 137 funeral homes listed in the Funeral Directors Association of New Zealand (2018) were emailed about participating in "a study on grave goods." The email outlined the aims of the study, explained that directors would be asked to "note down over the next couple of months how many of your clients are buried (or otherwise disposed of) with grave goods, and what these objects are," and defined grave goods as "objects that people are buried with (or in the case of cremation, placed along with the body in the crematorium). Examples of such objects include photographs, drawings, mementos, pet ashes, or idiosyncratic objects of personal and/or symbolic significance to the individual."

Of those contacted, directors from 12 funeral homes indicated their willingness to participate in the study.[1] These directors then received more detailed instructions for documenting their memorial services for a 3-month period, which involved the recording of service type (burial or cremation), the gender, age, ethnicity, and religion of the deceased, and any grave goods that were present "in as much detail as you feel comfortable" (excluding ordinary clothing worn by the deceased). One week after data collection began, participating funeral homes were sent a reminder email inviting any questions or feedback. A second reminder, at the end of the data collection period, asked if more time (another month) was required. All data were collected within 4 months.

2.2. Results

Funeral directors provided data from 357 funerals (168 females; 189 males), most of which were cremations (79%) with the remainder burials. The mean age of the deceased was 75.1 years ($SD = 17.8$) and ranged from 0 (fetuses that had not been carried to term) to 104. A majority (81%) identified as "New Zealand European" or another European ethnicity; 4.6% Maori; 1.4% undeclared; and the remainder included other ethnicities (some mixed).

A majority of funerals (216, or 61%) included grave goods of between 1 and 18 items (mean = 2.76; $SD = 2.37$; total = 596), with female decedents being more likely than males to include such items, $\chi^2(1) = 5.04$, $p = .025$, OR = 1.63. Age of the deceased ($p = .80$) and service type[2] ($p = .15$) were not associated with grave-good deposition.

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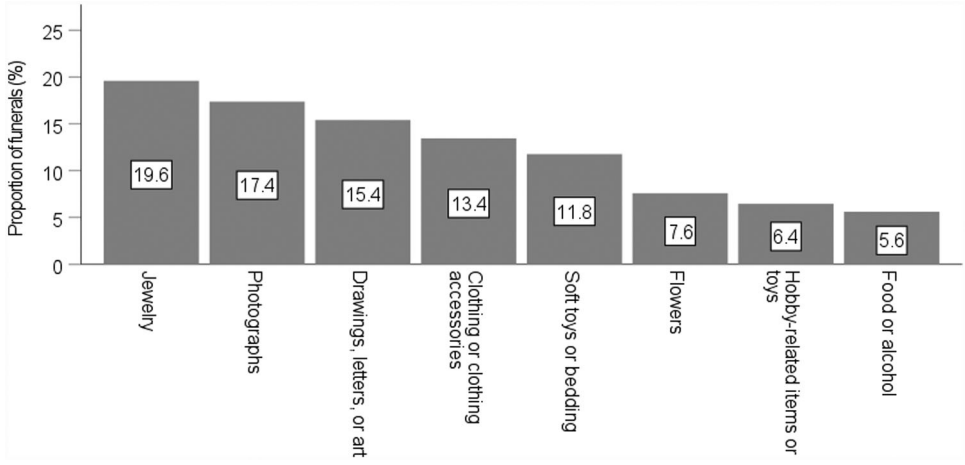


Fig. 1. The proportion (%) of funerals that included each type of grave good.

The 596 grave goods were coded by one of the authors and two hypothesis-blind research assistants in terms of whether they were best categorized as: *jewelry* (136, 22.8%); *drawings, letters, or art* (70, 11.7%); *photographs* (67, 11.2%); *clothing or clothing accessories* (63, 10.6%); *soft toys or bedding* (54, 9.1%); *hobby-related items or toys* (30, 5.0%); *flowers* (29, 4.9%), or *food or alcohol* (23, 3.9%). These were the eight largest categories to emerge from the data. The remaining 124 grave goods (20.8%) were coded as "other." Coders agreed 95% of the time and all disagreements were subsequently resolved through discussion or by majority decision.

Of the 216 funerals that included grave goods, 63% included more than one item, and 51% included more than one type of item. For each type of grave good, Fig. 1 shows the proportion of funerals that included at least one grave good of that type.

The religious affiliations of the deceased were not always recorded (42% were either unknown to the funeral home or not described in sufficient detail). The 27% of funerals that were nonreligious were compared with the 31% that were religious. The proportion of funerals that included grave goods did not differ by religiosity ($p = .25$).

2.3. Discussion

Consistent with previous research concerning Western, contemporary funerals (Elliot, 1990; Harper, 2012), a large percentage of New Zealand funerals (61%) included grave goods, suggesting that the practice remains popular, even in a relatively unreligious society.[3] The most common grave goods were jewelry, photographs, and "drawings, letters, or art," each of which may have socioemotional significance (e.g., wedding rings, family photos, and messages from family). The next most common types of grave goods were clothing and soft toys or bedding, categories that may carry personal significance (e.g., a military uniform or a childhood teddy). Following this were flowers, hobby-items or toys, and food or alcohol, items that may reflect preferences or customs (e.g., a favorite flower, book, or beverage).

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Thus, emotionality, particularly in relational contexts, appears to partially explain the deposition of grave goods in this study, which may also help to account for why funerals for female decedents included more grave goods than funerals for males. In most cultures, women express grief and mourning in more palpable and pronounced ways than men (Rosenblatt, Walsh, & Jackson, 1976). However, it is unclear if the gender difference is due to a greater number of female bereaved at the funerals of female deceased, to the bereaved exhibiting more emotion when the deceased is female, or to more prefuneral requests from females to be buried with grave goods. More generally, the extent to which grave goods are a bereaved-driven or deceased-driven practice is unclear and requires further investigation.

Curiously, burials were not significantly more likely than cremations to include grave goods. People may consider grave goods to be transported and manifested incorporeally with the soul of the deceased, similar to the Chinese tradition of burning paper money and models of precious objects before or at the funeral (such that the spirit will have them in the afterlife; Härke, 2014). Incorporeal objects would be consistent with studies of people's implicit reasoning about the afterlife, which have found that emotional and epistemic states are more likely to be attributed to the dead than are "psychobiological" states, and thus dead agents may be seen as having less physical (i.e., bodily) interactions with objects (see Bering, 2006).

Although the purpose of grave goods remains unclear from this naturalistic study, of particular importance is our finding that religious and nonreligious funerals did not differ in grave-good practices. This preliminary result warrants further investigation regarding whether that of the latter are motivated by implicit reasoning about an afterlife, but it does not support our hypothesis that religious funerals would include more grave goods (for afterlife use). However, the current study was limited in that a nonreligious person may still believe in an afterlife and a religious person may not (e.g., if manifesting extrinsic rather than intrinsic religiosity). Furthermore, it is somewhat unclear whether funeral directors were recording the religiosity of the deceased or of the bereaved.[4] A religious service may reflect the beliefs of either, both, or perhaps neither (e.g., if it is merely a nod to tradition). To disentangle the possible psychological factors involved in wanting/leaving grave goods (e.g., afterlife use, magical contagion, social signaling, and comfort of bereaved), we, therefore, adopted an experimental design that also included a more precise measure of explicit afterlife beliefs.

3. Study 2

In this study, we examined U.S. participants' preferences for wanting (or not wanting) grave goods at their own funerals, their stated reasons for deciding to include any such objects with their remains, and their religiosity and afterlife beliefs. In doing so, we hoped to exclude the motivations and influence of the bereaved. Regarding the core question of whether grave goods are motivated primarily by afterlife reasoning, even among those who explicitly reject the existence of an afterlife, we made two main predictions. First, we hypothesized that religiosity and explicit afterlife beliefs would make the need to provision oneself after death more salient (i.e., the "afterlife use" theory) and, thus, that participants with these beliefs would have a stronger preference for grave goods than participants without. Second, based

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on previous studies indicating that afterlife reasoning is a "cognitive default" (Bering, 2006), we predicted that extinctivists' stated reasons for wanting grave goods would not differ from afterlife believers in being motivated primarily by implicit assumptions about psychological continuity after death.

3.1. Method

3.1.1. Participants

A power analysis assuming a medium effect size for our main hypothesis (of an effect of afterlife belief on grave-good preference) indicated that the study required at least 450 participants to obtain power of .80 when α is set to .05. Separate groups of extinctivists (who agreed with the statement that "personal consciousness ceases permanently at death") and nonextinctivists were included in the study. Extinctivists were drawn from a pool of previously screened participants and paid US$2 in exchange for their participation. Nonextinctivists were drawn from a general sample and paid US$.50.[5] Four hundred and sixty-eight U.S. residents (193 females, 274 males, and one identifying as another gender) were ultimately recruited from Amazon's Mechanical Turk.[6] Their mean age was 38.5 years ($SD = 12.2$); 188 identified with a religion (of which 89% were Christian); 271 identified as "agnostic," "atheist," or "none"; and nine identified as "other" (e.g., spiritual). Regarding afterlife beliefs, of the total sample, 198 were extinctivists, 144 were afterlife believers, and 126 were agnostic (uncertain).[7]

3.1.2. Materials

Afterlife beliefs were measured with the Afterlife Beliefs Scale (ABS; Thalbourne, 1996), which asks participants to select one of six statements that "corresponds most closely to your own opinion about the possibility of life after death." The six statements describe distinct types of afterlife belief: *extinctivist* (consciousness "ceases permanently"); *immortalist* (consciousness "continues to exist forever"); *agnostic* ("uncertain as to what happens to the conscious personality at the death of the physical body"); *reincarnationist* ("reincarnated into a new body"); *eclectic* (a combination of immortalist and reincarnationist beliefs); and *other believer* (there is an afterlife "but I am completely unsure" what it is).

3.1.3. Procedure

The survey, described as being about "burial preferences," was administered in the Qualtrics survey environment and completed online. After providing written informed consent, participants were asked to "Imagine that you live in a society where all dead bodies must be interred (buried) in a grave at the cemetery. Many people are buried with things (e.g., family photos and a beloved pet's ashes). In preparing for your death, are there any particular items or objects that YOU would like to be buried with?" If participants responded "yes," they were asked to list these items and, on the next screen, to "explain why you want to be buried with the items you listed," typing their responses into text boxes. If participants responded "no," these follow-up responses were not requested. Prior to debriefing, participants completed demographic questions (including religious affiliation) and the ABS.

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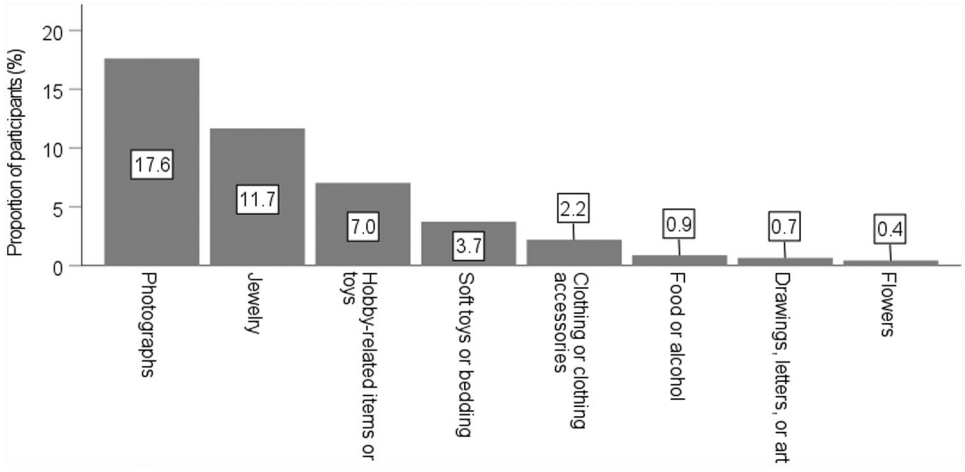


Fig. 2. The proportion (%) of participants who wanted each type of grave good.

3.2. Results

Fourteen participants did not follow instructions and were excluded from the analyses. A majority of participants (261, or 57%) did not want to be buried with grave goods. The 193 participants who wanted grave goods each listed between 1 and 5 items (mean = 1.68; *SD* = 0.90; total = 325), with 45% wanting more than one object, and 35% wanting more than one type of object. Grave-good types were coded within the categories established in Study 1, with coders initially agreeing 92% of the time, and disagreements subsequently resolved via discussion. For each, Fig. 2 shows the proportion of participants who wanted to be buried with at least one grave good of that type. Female participants were more likely to want grave goods than males, $\chi^2(1) = 5.66$, $p = .02$, OR = 1.58. Age was not associated with preference for grave goods ($p = .70$).

3.2.1. Religiosity and afterlife beliefs

Excluding participants whose religious affiliation was described as "other" (leaving 178 religious; 267 nonreligious), the proportion of participants who wanted to be buried with grave goods did not differ by religiosity ($p = .31$). The proportion of participants who wanted to be buried with grave goods differed by afterlife belief, $\chi^2(1) = 12.88$, $p < .001$, OR = 2.26, with a greater proportion of afterlife believers (immortalists, reincarnationists, eclectics, and "other") than extinctivists wanting such objects (54% vs. 34%). This result was qualitatively unchanged when agnostics were included in either group.

3.2.2. Reasons for wanting grave goods

The reasons participants gave for wanting to be buried with grave goods were coded by two of the authors (TS and RH) and a research assistant[8] in terms of whether they provided evidence of afterlife reasoning. Consistent with research about the cognitive states that people attribute to the dead (Bering, 2002; Bering & Bjorklund, 2004; Bering et al., 2005a;

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Georgiadou & Pnevmatikos, 2019; Misailidi & Kornilaki, 2015), a statement was coded as including "afterlife reasoning" if it mentioned or alluded to a postdeath self, such as by reference to postdeath knowledge or desires, or to continued existence at a location or future point in time. Examples of such statements included: "*It would just feel like they are with me*"; "*This blanket is like my best friend and I wouldn't feel right if I wasn't buried with her*"; "*I love my daughters so much I never want to be without them even after I'm gone*," and; "*I love them all very much. Hopefully we will all be together in the afterlife as well*."[9] By contrast, examples of statements without afterlife reasoning included: "*I would like to be buried with this because it is the most sentimental item I own*"; "*These items represent what has been important in my life*"; and "*The files are a summary of my life and might function as an interesting time capsule*." All three coders agreed 91% of the time, with a majority decision used for disagreements. Two participants were excluded for not providing a reason.

Overall, 46% of participants who wanted grave goods gave reasons that provided evidence of afterlife reasoning. The proportions of extinctivists (47%) and afterlife believers (50%) did not significantly differ in the extent to which their reasons demonstrated afterlife reasoning, $\chi^2(1) = .13$, $p = .72$, OR = 1.13. This result was qualitatively unchanged by the addition of agnostics to either group. Afterlife reasoning was also not associated with religiosity ($p = .41$), gender ($p = .76$), or age ($p = .81$).

3.3. Discussion

Consistent with Study 1 and previous research concerning Western, contemporary funerals (Elliot, 1990; Harper, 2012), many U.S. participants wanted to be buried with grave goods (43%), although the normative language in our survey instructions may have exaggerated the true number.[10] As in Study 1, socioemotional goods were the most common (jewelry, photographs), followed by personally significant goods (soft toys or bedding, clothing), and, finally, by preferences or customs (food or alcohol, flowers). However, "drawings, letters, or art" was a socioemotional good that was infrequently listed, while hobby-items and toys was a personal preference that was frequently listed. Furthermore, the use of "photographs" as an example in the instructions may have biased the results.

Because Study 2 reflected the preferences of the individual being "buried" but not those grieving them, the reduced preference for grave goods (Study 1 = 61%; Study 2 = 43%) and the smaller mean number of grave goods (Study 1 = 2.76; Study 2 = 1.68) might suggest that a large proportion of the grave goods in Study 1 were left by the bereaved without the deceased having requested them. This may also account for the large drop in "drawings, letters, or art" (Study 1 = 15%; Study 2 = 1%). These items (e.g., letters to the deceased) appear to be for the benefit of the bereaved. The increase in hobby-related items (compared to Study 1) may be related to such items being relatively popular among the large number of extinctivists in the present study (see Supplementary Materials). However, cultural differences between NZ and the United States, or "real-world vs. laboratory" differences (such as in attention/effort), may have also influenced these effects.

The fact that the majority (57%) of participants did not want grave goods can perhaps be understood by considering that grave-good practices are only a proxy for afterlife reasoning

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(e.g., one can believe in an afterlife without developing a need to provision oneself after death, and physical grave goods may not be the ideal provision for all participants). The result may also reflect a variety of factors, including a lack of engagement with the study, an inability to think of any objects, and perhaps extinctivist reasoning. In this latter context, especially, it is also important to highlight a core theme of dual-process models of cognition, which is that explicit "System 2" thinking tends to override implicit "System 1" intuitions (Evans, 2003). For example, such a pattern might occur for participants who reasoned that they would have everything they could possibly want or need in heaven, that they will be in a state of "soul sleep" until their physical body is resurrected and have no need for such items during this period, or that grave goods are inconsistent with other religious or secular traditions. These reasons for *not* engaging in grave-good practices, which might override participants' implicit beliefs, were investigated in our next study.

In the present study, grave goods were more common for female than male participants, which was consistent with Study 1. With no bereaved-driven contribution, this result suggests that there may be more prefuneral requests from females than males to be buried with grave goods. Similarly, as in Study 1, religious and nonreligious participants did not differ in their preference for grave goods, suggesting that implicit afterlife beliefs might drive grave-good practices without any explicit motivation. However, a more appropriate measure of explicit afterlife beliefs did predict a preference for grave goods, with extinctivists showing less of a preference. This correlation between explicit afterlife beliefs and grave-good practices, regardless of religiosity, is consistent with the "afterlife use" theory and suggests that other aspects of religiosity (e.g., ritual behavior and social signaling) are less salient to the practice.

Crucially, about one-third of extinctivists still wanted grave goods and, as with afterlife believers, about half of these participants referenced some form of postmortem existence, contradicting their explicit rejections of personal consciousness surviving death. The latter result suggests that intuitive afterlife reasoning is a common motivation for grave-good practices and, although the hypothesis that it is the primary factor could not be supported conclusively (i.e., the motivation was unclear for half of our participants), the result may be conservative, since responses that were indicative of afterlife reasoning were significantly longer ($M = 126$ characters) than nonindicative responses ($M = 99$ characters; $p = .03$), hinting that some participants might have indicated afterlife beliefs had they elaborated on their reasons. Alternatively, participants may have relied on colloquialisms, figurative speech, and/or found it difficult to express their reasons using language that does *not* imply continuing consciousness, and they might be less likely to do so if they were made aware of alternative reasons. We, therefore, sought to address these issues in our next study.

4. Study 3

In this study, the bereaved-driven contribution to grave-good practices was examined by asking participants to imagine the future funeral of a loved one. To more systematically understand the various motivations for leaving grave goods, participants were asked to select from

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among an extensive list of possible motivations drawn from previous literature, or alternatively to explain why they would *not* leave grave goods. They were also asked about their emotionality, closeness to the hypothetical deceased, and sensitivity to magical contagion (i.e., whether grave goods contain the deceased's "essence"; see Introduction), and about the value of each grave good (sentimental, economic, etc.), whether they would leave the object if no one were aware they had done so (social signaling), and the comfort it would bring.

Consistent with Study 2, we predicted that jewelry and photographs would be popular grave goods, owing to their sentimental, emotional, and relationship value, that extinctivists and afterlife believers would exhibit afterlife reasoning to an equivalent extent when explaining this behavior, and that the latter would prefer grave goods to a greater extent, explicitly referencing their afterlife use. Furthermore, given the pattern of findings thus far, we anticipated that the primary reason for *not* leaving grave goods would be something other than extinctivist reasoning (e.g., cultural restrictions, the deceased's preferences, or not wanting to cremate the objects).

Consistent with our core research question, we predicted that alternative motivations for leaving grave goods would be less important than afterlife reasoning. Specifically, we predicted that magical contagion beliefs would motivate some participants and be statistically explained by a need to ritually "complete" the person at death (i.e., incorporating with the body all objects that contain their unique essence; Elliot, 1990), but would also be a reason to not leave grave goods (e.g., keeping objects to maintain connections to the deceased; Gibson, 2010). Finally, we predicted that a personal need for emotional comfort would also motivate some participants (perhaps supported by emotionality, closeness to the deceased, and some demographic variables; Elliot, 1990; Layne, 2000), but that other social and reputational motives would be less important in contemporary, Western contexts (e.g., social signaling; Härke, 2014).

4.1. Method

4.1.1. Participants

Based on the power analysis reported in Study 2, this study also required at least 450 participants to obtain power of .80 when α is set to .05. Four hundred and sixty-six volunteers were ultimately recruited[11] (220 females, 246 males) from Mechanical Turk and paid US$2. All were native English-speaking U.S. nationals currently living in the United States. Their average age was 40.0 years ($SD = 12.4$); 259 identified with a religion (of which 88% were Christian); 196 identified as "agnostic," "atheist," or "none"; and 11 identified as "other" (e.g., spiritual). Of the total sample, 111 were extinctivists; 210 were afterlife believers; and 145 were agnostic (uncertain) about whether there is an afterlife.

4.1.2. Materials

A Contagion Sensitivity Scale (CSS; Fedotova, 2013; $\alpha = .84$) was used to measure magical contagion beliefs. Participants rated from 1 (strongly disagree) to 9 (strongly agree) their agreement with 22 statements that describe four kinds of contagion beliefs: (1) *negative spiritual contagion*, for example, "I would not want to sit in a chair if I knew that a day before

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a criminal had sat in that chair," (2) *positive celebrity contagion*, for example, "I would like to touch a pen that was used to sign an important treaty," (3) *positive personal contagion*, for example, "I own and value some objects because they once belonged to someone I love," and (4) *negative physical contagion*, for example, "I prefer to wash my hands pretty soon after shaking someone's hand."

Participants' reasons for leaving grave goods were examined by asking them to choose (all that apply) from 26 statements that expressed 12 types of grave-good motivation (Table 1).[12] These general types of motivation were derived following a thorough review of related literature (see Introduction), while the individual statements were based on participant responses or author descriptions from this body of extant research and assigned to motivation categories based on how these authors had interpreted their findings.

An *explicit afterlife belief* motivation was represented by three statements about afterlife use. The statements accounted for grave goods that are "used in the next world" (Elliot, 1990, p. 608), used on the deceased's "unfamiliar journey" (Harper, 2012, p. 52), or used to "prevent their return" (Härke, 2014, p. 51). An *implicit afterlife belief* motivation was represented by three statements that described the dead as being able to know, feel, or remember (Bering & Bjorklund, 2004), as well as by six statements that described other motivations, but which implied postmortem consciousness.[13]

A *magical contagion* motivation was represented by seven statements that treated grave goods as a physical remnant or "part" of the deceased (e.g., as for some consumer objects and celebrity memorabilia; Huang et al., 2017; Newman et al., 2011). The notion that objects can be imbued or polluted with the deceased's essence (Härke, 2014; Klevnäs, 2016) also underlies the view of grave goods as *inalienable property* (objects that "could not be inherited, sold off or given away"; Härke, 2014, p. 45), *ritual completion* (the deceased must be interred "whole" or complete with all necessary limbs and objects; Elliot 1990, p. 606) and *reminder theory* (objects must be "out of sight so as to not remind them of their loss"; Elliot, 1990, p. 609).

Various forms of emotional comfort were also considered. A *personal comfort* motivation was represented by two statements that described doing "one final thing" for the loved one (Elliot, 1990, p. 609) or "receiving comfort in the gifting process" (Drenten et al., 2017, p. 450). A *closure* motivation was represented by a statement about saying goodbye "to put final touches on a relationship" (Elliot, 1990, p. 611). The *other comfort* motivation was represented by three statements about comforting the family of the deceased (i.e., the "sentiment of relatives"; Elliot, 1990, p. 606).

Finally, various social motives were considered. The *social contract* motivation was represented by two statements about honoring what the deceased "wished to be buried [with]" (Harper, 2012, p. 48). A *deceased's status* motivation was represented by two statements about honoring the dead with goods that "reflect the social status of the deceased" (Härke, 2014, p. 47). A *social signaling* motivation was represented by five statements for which there may be a reputational benefit to the bereaved from either a show of wealth (i.e., "potlatch"; Härke, 2014, p. 46), a show of emotional status (i.e., "a statement to the world that I remember my loved one"; Drenten et al., 2017, p. 423), conformity with religious or other traditions (Härke, 2014, pp. 49–50), or general conformity with other people's grave-good

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Table 1
Statements to explain grave-good deposition, the types of motivation they represent, and the proportion (%) of participants who endorsed them

Statement	Motivation (ALB = afterlife belief)	*N* (%)
Because it is part of who they were in life.	Magical contagion	62.4
To say goodbye and put the final touches on our relationship.	Closure	57.1
To comfort myself by doing one final thing for them.	Personal comfort/implicit ALB	48.7
So that a part of me and/or the family will be with them.	Implicit ALB/magical contagion	48.1
So that they will know I loved them.	Implicit ALB	46.6
So that things that are part of who they are will be with them.	Implicit ALB/magical contagion	43.9
So that they will be at rest.	Implicit ALB	29.1
So that they will be complete.	Implicit ALB/magical contagion/ritual completion	25.9
So that they will have it in the afterlife.	Explicit ALB	24.9
To comfort myself by giving a gift even though I do not think it will be received.	Personal comfort	23.3
To honor the deceased's wishes so that they will be at rest.	Social contract/implicit ALB	23.3
To show everyone how much I cared for the deceased.	Social signaling (emotional status)	22.8
To comfort the family so they will know that the deceased was loved and will be remembered.	Other comfort	21.2
Because the ceremony would be incomplete without including all the things that made up who they were in life.	Magical contagion/ritual completion	20.1
So that they will not forget me.	Implicit ALB	18.5
Because you cannot give away or sell something that was such a big part of who a person was in life.	Magical contagion/inalienable property	16.9
To honor the deceased's wishes so that the family will be comforted.	Other comfort/social contract	16.9
To help them get to the afterlife.	Explicit ALB	11.6
To honor the rank or status of the deceased so that they will be at rest.	Deceased's status/implicit ALB	11.1
To show everyone that I respect the religious nature of the ceremony.	Social signaling (religious)	10.1
To honor the rank or status of the deceased so that the family will be comforted.	Deceased's status/other comfort	10.1
To show everyone that I respect the traditions of the ceremony.	Social signaling (other tradition, e.g., military)	9.0
Because other people will be leaving things, so it will be expected of me too.	Social signaling (conformity)	6.9
So that it is out of sight and will not remind me/us of our loss.	Magical contagion/reminder theory	4.2
So that they will not want to return to our world.	Explicit ALB	2.6
To show everyone that I have made an expensive contribution to the ceremony.	Social signaling (wealth)	1.1

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practices. A summary of the candidate motivations, the statements representing them, and the proportion of participants endorsing those statements appears in Table 1.

4.1.3. Procedure

The survey, described as being about "contemplating the funeral of a loved one," was administered in the Qualtrics survey environment and completed online. Participants were informed that the questions might be distressing. After providing written informed consent, participants were asked to "think about a family member or close friend who is currently still alive but may pass away in your lifetime, and whose funeral you would attend" and to provide this person's gender, current age, and expected service type (burial or cremation). Participants also rated from 1 (not at all) to 7 (extremely) "how close are you to this person," "how upset would you be by this person's death," and "how visibly upset do you think you would be at the funeral."

Participants were then asked if they would leave grave goods with the deceased.[14] If they responded "yes," they were asked to list up to 10 items within separate text boxes. They were then reminded of each item (on separate screens) and responded "yes" or "no" to questions designed to measure social signaling and emotional comfort: "Would you still place this object with the deceased if no other living person would ever know you had done so?" and "Would placing this object with the deceased help you to grieve/bring you comfort?" Participants were also asked "What kind of value does this object have?" and to select all that apply from: *sentimental value* (it symbolizes or brings up memories), *emotional value* (it symbolizes or brings up emotions), *relationship value* (it represents a relationship with the deceased), *functional value* (it is useful), *economic value* (it is expensive), *social or community value* (it would be appreciated by people at the funeral or in the community), *cultural or traditional value* (it would be appreciated by people from my culture or traditions), *other value*, and *no value*.[15] Participants were then shown all of their items and were asked to select all statements (from Table 1) that describe their reasons for leaving them. If participants indicated they would not leave grave goods, these responses were not requested, and they were instead asked to explain (in a text box) why they would not.

All participants were then asked to "imagine that the funeral is over" and to answer a question designed to measure the need for ritual completion: "How upset would you be if you found an object later on that you thought should have been placed with [the decedent], but for some reason was not?" by rating their response from 1 (not at all) to 7 (extremely). Next, participants completed the ABS (as in Study 2) and, if they indicated any afterlife belief, were asked "Do you think the person whose funeral you have been thinking about will have everything they want/need after death?" by responding "yes" or "no" (participants who left grave goods were told to imagine they forgot to leave them). Prior to debriefing, participants completed the CSS and demographic questions about themselves (including religious affiliation).

4.2. Results

Certain participants were excluded from specific analyses for lack of attention based on a priori, task-specific time constraints that reflected the text length of items and how long it

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Table 2
A comparison of grave-good practices of the to-be-deceased (Study 2) and to-be-bereaved (Study 3)

Grave good	Study 2 (%)	Study 3 (%)	$\chi^2(1)$	p
Photographs	17.6	19.2	.35	.55
Drawings, letters, or art	0.7	5.1	16.07	< .001
Clothing (and accessories)	2.2	8.9	19.41	< .001
Jewelry	11.7	16.7	4.69	.03
Flowers	0.4	8.9	36.50	< .001
Soft toys or bedding	3.7	4.0	.04	.84
Food or alcohol	0.9	2.4	3.40	.07
Hobby-related items or toys	7.0	10.7	3.71	.05
"Other"	17.0	17.8	.12	.73

would take to read them.[16] Specifically, five participants were excluded from analyses involving the information they provided about the deceased (< 2 s per question), 29 from analyses involving the statements in Table 1 (< 0.5 s per item), 30 from analyses involving the ABS (< 1 s per item), and 30 from analyses involving the CSS (< 2 s per item). Additionally, 17 participants were excluded from all analyses involving the grave goods they listed because they appeared not to follow or understand the instructions.

A majority of participants (248, or 55%) indicated that they would not leave objects with the deceased. Of the 201 participants who indicated that they would leave objects, each listed between 1 and 10 items (mean = 2.60; SD = 1.64; total = 522), with 74% leaving more than one item, and 66% leaving more than one type. These objects were categorized as in Studies 1 and 2, with coders agreeing 88% of the time (a majority decision resolved disagreements). Table 2 shows the proportion (%) of participants who would leave each type of grave good compared with the results from Study 2. The comparison suggests large bereaved-driven contributions to the deposition of "drawings, letters, or art," "clothing," and "flowers."

Grave-good deposition was not associated with participant gender (p = .59), deceased gender (p = .23), or participant age (p = .48), but the age of the deceased was younger when grave goods were deposited (M = 55.93; SD = 19.84) than when they were not (M = 65.62; SD = 17.88), $t(432.70) = 5.47$, $p < .001$, r = .25. Participants who indicated they would leave grave goods would also be more *visibly* upset at the funeral (M = 5.42; SD = 1.36) than those who responded they would not (M = 5.07; SD = 1.71), $t(456.09) = 2.43$, p = .02, r = .11, but there was no association with being generally upset (p = .98), closeness with the deceased (p = .79), or service type (p = .13).

4.2.1. Grave-good value

For the 522 grave goods, a social signaling hypothesis was not supported, as 96% would still be left with the deceased "if no other living person would ever know." A comfort hypothesis was supported, as the deposition was deemed comforting for 92%.[17] In total, 68% of grave goods were reported to have sentimental value, 59% emotional, 52% relationship, 21% cultural/traditional, 17% social/community, 14% functional, 11% economic, 2% other, and 0% none. A series of χ^2 tests determined whether there were differences between the nine

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grave-good categories on these measures, which are reported in full in the Supplementary Materials. Bonferroni-corrected pairwise comparisons indicated that "drawings, letters, or art" had more emotional and relationship value, photographs had more relationship value, jewelry had more economic value, and "other" items had more cultural/traditional value (about a quarter of "other" items were Bibles and rosaries[18]) than some other categories of grave good, although most comparisons were not significant.

4.2.2. Religiosity

Excluding participants whose religious affiliation was described as "other" (leaving 259 religious; 196 nonreligious), religious participants were more likely to leave grave goods than the nonreligious, $\chi^2(1) = 10.29$, $p = .001$, OR = 1.85. Compared to the nonreligious, religious participants were more likely to leave grave goods: to socially signal $\chi^2(1) = 13.08$, $p < .001$, OR[19] = 13.42, with functional value, $\chi^2(1) = 14.03$, $p < .001$, OR = 3.40, economic value, $\chi^2(1) = 15.94$, $p < .001$, OR = 5.03, social or community value, $\chi^2(1) = 13.77$, $p < .001$, OR = 2.95, and cultural or traditional value, $\chi^2(1) = 8.21$, $p = .004$, OR = 2.01. Nonreligious participants were more likely to leave grave goods with sentimental value, $\chi^2(1) = 22.10$, $p < .001$, OR = 2.75, and relationship value, $\chi^2(1) = 4.34$, $p = .04$, OR = 1.48.

4.2.3. Afterlife belief

A binomial test revealed that most afterlife believers (88%) thought that the deceased would have everything they want/need after death rather than not ($p < .001$), but afterlife believers (50%) were still more likely than extinctivists (32%) to leave grave goods, $\chi^2(1) = 8.59$, $p = .003$, OR = 2.09.[20] Afterlife believers were more likely than extinctivists to report that leaving grave goods would bring them comfort, $\chi^2(1) = 6.64$, $p = .01$, OR = 5.56, but there was no association with social signaling ($p = .18$) or with grave goods with any particular type of value ($ps > .08$).

4.2.4. Contagion sensitivity and ritual completion

Participants who would leave grave goods had higher CSS scores[21] ($M = 101.62$; $SD = 29.40$) than participants who would not ($M = 88.06$; $SD = 33.96$), $t(430.72) = 4.46$, $p < .001$, $r = .21$, and scores did not vary across grave-good categories ($p = .64$). Females scored higher ($M = 101.94$; $SD = 29.86$) than males ($M = 86.73$; $SD = 33.56$), $t(433.20) = 5.01$, $p < .001$, $r = .23$, religious participants scored higher ($M = 101.76$; $SD = 33.35$) than nonreligious participants ($M = 84.03$; $SD = 32.07$), $t(423) = 5.74$, $p = < .001$, $r = .27$, and afterlife believers scored higher ($M = 102.07$; $SD = 30.75$) than extinctivists ($M = 80.91$; $SD = 33.36$), $t(290) = 5.43$, $p < .001$, $r = .30$. Mediation models showed that the effect of religiosity on leaving grave goods was fully explained by CSS score (indirect: b = .22, 95% CI [.10, .37]; direct: b = .25, 95% CI [–.15, .66]), but that the effect of afterlife belief on leaving grave goods was not explained by CSS score (indirect: b = .13, 95% CI [–.02, .32]; direct: b = .71, 95% CI [.17, 1.24]).

Ritual completion (the emotional importance of including all necessary grave goods) was associated with leaving grave goods, $t(462.05) = 12.04$, $p < .001$, $r = .49$, and was correlated with CSS score, $p < .001$, $r = .33$, even when only including participants who would leave

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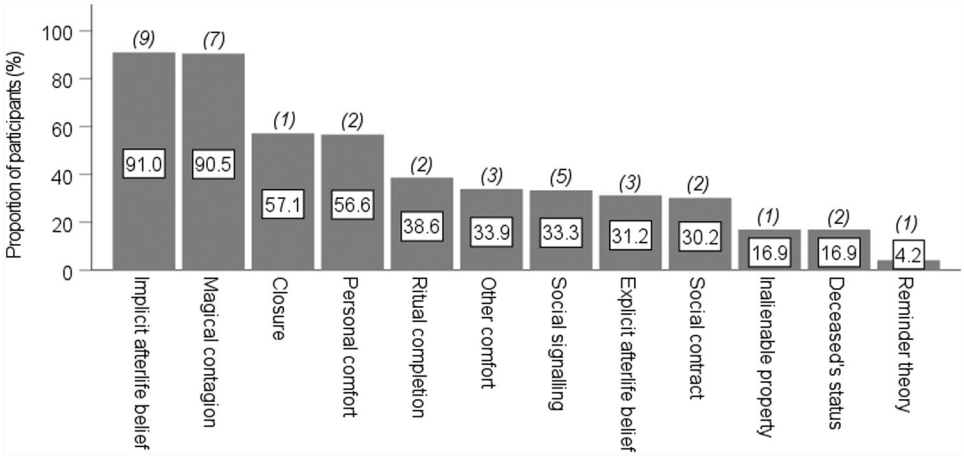


Fig. 3. The proportion (%) of participants who selected at least one of the statements in these motivation categories. The numbers in parentheses represent the number of statements from Table 1 in each category.

grave goods, $p = .008$, $r = .19$. A mediation model showed that the effect of CSS score on leaving grave goods was fully explained by ritual completion (indirect: b = .011, 95% CI [.007, .016]; direct: b = .005, 95% CI [–.002, .012]).

4.2.5. Reasons for leaving grave goods

The 26 statements in Table 1 were selected 1165 times by 189 participants ($M = 6.16$; $SD = 3.71$; range = 0–22)[22] and were recoded into 12 categories representing the types of motivation for leaving grave goods described in Table 1. The proportion of participants who selected *at least one* statement in these categories is shown in Fig. 3.

Conceptual overlap between motivation categories and (consequently) the many statements in the top two categories in Fig. 3 may have contributed to their popularity. Categories were, therefore, created that included a more conservative and literal set of statements with no conceptual overlap. There were three "implicit afterlife belief" statements that did not overlap (see Table 1), and 59% of participants endorsed at least one of these. Only one "magical contagion" statement did not overlap, endorsed by 62%. Thus, these motivations remained the most popular, regardless of statement frequency.[23]

Consistent with the "afterlife use" theory, statements about provisioning the dead (i.e., the "explicit afterlife belief" motivation) were endorsed more by afterlife believers than by extinctivists, $\chi^2(1) = 5.51$, $p = .02$, OR = 3.68. Importantly, however, these groups did not differ in being motivated by implicit afterlife belief, $\chi^2(1) = 1.34$, $p = .36$, OR = 2.45, and this result was qualitatively unchanged when the more conservative category (with three statements) was considered, $\chi^2(1) = .04$, $p = .84$, OR = 1.09. The groups also did not differ for any other motivation categories ($ps > .15$). Religious and nonreligious groups were also compared. Religious participants were more likely than nonreligious participants to be motivated by "other comfort," $\chi^2(1) = 6.21$, $p = .01$, OR = 2.30, social signaling, $\chi^2(1) = 4.23$, $p = .04$, OR = 1.98, and the deceased's status, $\chi^2(1) = 8.08$, $p = .004$, OR = 3.99 (other $ps > .17$).

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Table 3
Participants' reasons for *not* wanting to leave grave goods and the proportion (%) of participants in each category

Category	*N* (%)	Example response
Bereaved's personality	35.9	"I am not against placing objects in a casket or on a person. It is just something that I prefer not to partake in. Personal preference."
Deceased's wishes	17.7	"This person has stated that he doesn't want anything placed in his casket when he is buried."
Deceased's personality	16.5	"There are not many material objects that are very special to them."
Magical contagion	16.5	"I believe they are more valuable to the people that are alive so they can remember the person."
Cannot think of anything	13.7	"I don't really have an object that I would associate with them."
Not customary or traditional	10.1	"It's not something people in my family have ever done."
Extinctivism	8.9	"I don't really believe in life after death or anything like that. Also the object would end up sitting with a rotted carcass."
Destruction in cremation	6.9	"Why would I have something cremated with them? Just so it could be destroyed? That doesn't make any sense to me and it wouldn't make it any less painful."
Inappropriate or disrespectful	3.6	"I would think it would be disrespectful if I just put anything in there."

4.2.6. Reasons for not leaving grave goods

The 322 reasons given by 248 participants for *not* wanting to leave grave goods were coded by two of the authors (TS and RH) and a hypothesis-naïve research assistant within the nine categories summarized in Table 3 (the complete coding scheme is in the Supplementary Materials). All three coders agreed 77% of the time, with a majority (2-1) decision used to resolve disagreements. Three participants were excluded for not following instructions. The use of explanation did not differ as a function of religiosity or of afterlife belief.

4.3. Discussion

In this study, 45% of U.S. participants indicated that they would bury or cremate grave goods with a deceased loved one, similar to the percentage wanting to be buried with grave goods in Study 2 (43%). Collectively, these bereaved- and deceased-driven contributions may account for the numbers observed at real funerals. A comparison between Studies 2 and 3 (Table 2) identified large bereaved-driven contributions to the deposition of "drawings, letters, or art" (DLA), clothing, and flowers, which may explain their prevalence in Study 1, although DLA may be especially common at NZ funerals. Gender did not predict grave-good

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deposition, suggesting that prior gender effects (females were more likely to have grave goods in Studies 1 and 2) are deceased-driven (e.g., prefuneral arrangements, in a will, etc.).

Despite finding that grave goods are largely chosen for their sentimental, emotional, and relationship value, rather than their functional, economic, social, or cultural value, there was more uniformity in this value across grave-good categories than expected (see Supplementary Materials). In other words, a special piece of clothing may have as much sentimental value as a piece of jewelry. The prevalence of jewelry and photographs in all three studies thus suggests that these items are more likely to hold sentimental, emotional, and relationship value compared to other types of objects, but that, when other objects *are* deposited, it is because they hold similar value (e.g., a baseball hat from a memorable game).

Inconsistent with Studies 1 and 2, religious participants were more likely to leave grave goods, which our results suggest is a bereaved-driven effect. Religious participants were significantly more likely to indicate that they would be visibly upset at the funeral than were the nonreligious, to leave "other" items (that included many Bibles and rosaries) with social/community value or cultural/traditional value, and to be motivated more by desires to comfort the family, to socially signal, and to honor the status of the deceased.[24] The bereaved-focused design of the present study thus appeared to highlight the extrinsic social motivations of religious individuals to visibly apply religious values and traditions in ritual settings. Additionally, despite religious individuals not having a greater magical contagion motivation (Table 1) than the nonreligious, contagion sensitivity (CSS score) fully explained their greater grave-good deposition, which might suggest an implicit contagion effect behind these religious displays, or reverse causality (e.g., contagion sensitivity may influence religiosity rather than the converse).

Consistent with Study 2, and the "afterlife use" theory, afterlife believers left more grave goods (50%) than extinctivists (32%), with the present study providing support for our view that the former are motivated by their explicit beliefs to provision the dead (e.g., greater agreement with the statement "So that they will have it in the afterlife"). Explicit provisioning of the dead was also the only motivation category on which the two groups differed. The result may explain why grave-good deposition was more comforting for afterlife believers. However, most afterlife believers thought the afterlife provides for all the needs/wants of the deceased (e.g., the Christian paradisaical afterlife), which may reduce grave-good deposition relative to other cultural-historical contexts.

Importantly, even in the present study, where *implicit* afterlife belief statements (i.e., statements that by some necessity assume the dead still have the capacity to know, feel, or remember) were prewritten and presented alongside alternatives, extinctivists still endorsed these statements as reasons for their grave-good deposition as much as afterlife believers did, which suggests that such responses cannot be easily attributed to linguistic constraints or casual, off-the-cuff colloquialisms involving mental states after death. Overall, 91% of participants who chose to leave grave goods endorsed at least one of these statements (Fig. 3), suggesting the widespread use of afterlife reasoning, although the large number of statements that employed afterlife reasoning may have contributed to this result. When a more conservative selection of statements was considered (i.e., only literal and discrete descriptions of the dead being able to know, feel, or remember), endorsement remained high (59%), with extinctivists still not differing from afterlife believers.

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Although all of the motivations for leaving grave goods received some support, implicit afterlife belief (91%, e.g., "*So that they will know I loved them*"), magical contagion (90%, e.g., "*Because it is part of who they were in life*"), closure (57%, "*To say goodbye and put the final touches on our relationship*"), and personal comfort (57%, e.g., "*To comfort myself by doing one final thing for them*") were most common. Closure may be another form of personal comfort, in terms of attenuating grief, and 92% of grave goods were independently rated as helping "you to grieve/bring you comfort" (grave-good deposition was also associated with being visibly upset at the funeral and inversely related to the deceased's age). Conversely, given that 96% of grave goods would still be left with the deceased "if no other living person would ever know," social signaling (33%) and comforting others (34%) are unlikely to be primary motivations. Reminder theory (4%), inalienable property (17%), and ritual completion (39%) are each associated with magical contagion, with the latter being most important, while honoring a social contract (30%) obscures the deceased's motivation for wanting the object.

Consistent with these findings, contagion sensitivity (CSS score) was associated with grave-good deposition, and both variables were associated with ritual completion (the emotional importance of including all necessary grave goods), which fully explained the relationship, suggesting that including objects that are "part of" someone is intended to reconstruct the essence of the deceased as part of the funerary ritual. Although the CSS and the Table 1 statements both indicated that magical contagion is a common motivation for grave-good practices, the latter result should be regarded with some caution. It may be that describing grave goods as "part of" the deceased was meant figuratively by some participants. Although our review of the empirical literature (particularly Newman et al., 2011) suggests that a literal treatment of such statements is reasonable (i.e., that the grave good is thought to be imbued with the deceased's essence and/or that including it with the body is necessary to ensure the person's intactness at burial), we cannot rule out a figurative understanding, which does not necessarily imply a magical contagion motivation.

Some participants who did *not* leave grave goods also gave reasons related to magical contagion (17%). Overall, however, reasons for not leaving grave goods were diverse (Table 3), with most associated with the personality of the to-be-bereaved or the personality/wishes of the to-be-deceased, suggesting that there are many ways in which explicit beliefs, sentiments, and traditions might override intuitive afterlife reasoning (including 14% who simply could not think of a suitable object). In addition, 88% of explicit afterlife believers thought the deceased would already have all they need/want (e.g., in heaven), which may have contributed to many participants choosing not to leave grave goods. Extinctivism (i.e., "there is no point because the person no longer exists") was rarely given as a reason (9%), which was consistent with our finding that extinctivists are unlikely to override intuitive afterlife reasoning with their explicit beliefs when deciding to leave grave goods.

5. General discussion

In the present work, we asked if explicit afterlife beliefs motivate grave-good practices and whether such practices provide any evidence for intuitive afterlife reasoning (i.e., an implicit

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cognitive predisposition to envisage consciousness as surviving death). To test the role of these implicit afterlife beliefs, independent of explicit endorsement, we drew on the high populations of extinctivists/skeptics in Western countries. Across three studies, conducted in the United States and New Zealand, we contrasted people's explicit religiosity and stated afterlife beliefs with their grave-good practices at actual funerals or at their own (hypothetical) funerals or that of their loved ones. Based on findings from previous, related studies, we predicted that "afterlife use" would be the primary motivation for grave-good practices. Specifically, we hypothesized that explicit religious/afterlife beliefs motivate the practice, but that ascribing mental states to the dead also underlies many nonreligious/extinctivists' reasons for burying (or burning, in the case of cremations) objects with the dead.

In Study 1, 61% of actual NZ funerals were found to include grave goods, with socioemotional items, such as photographs, jewelry, and "drawings, letters, or art" (DLA) being the most common, followed by personally significant items, such as clothing and soft toys or bedding, and finally by items that reflected customs and preferences, such as flowers, hobby-related items, and food or alcohol. Funerals for females included more grave goods than for males, but it was unclear whether this effect was bereaved-driven or deceased-driven (e.g., directives in a will). Contrary to our prediction, religious and nonreligious funerals did not differ in their grave-good practices, suggesting that explicit beliefs do not play a role; however, it was unclear whose religiosity was reflected at the funeral, or whether religiosity could be considered a reliable proxy for afterlife beliefs. Nevertheless, the results from our naturalistic study on actual funerals warranted an investigation of whether intuitive afterlife reasoning motivates the grave-good practices of the nonreligious.

In Study 2, 43% of U.S. participants wanted to be buried with grave goods, which were of similar types and proportions to Study 1, although DLA less so, and "hobby-items and toys" more so. Consistent with Study 1, female participants wanted more grave goods than males, suggesting the effect is deceased-driven, while religious and nonreligious participants did not differ in preference for grave goods. Afterlife believers preferred grave goods more than did extinctivists, suggesting a role for explicit beliefs, while the reasons provided by afterlife believers and extinctivists suggested they were motivated equally (46% of the time) by intuitive afterlife reasoning (i.e., the assumption that the dead self would be an experiential agent).

In Study 3, 45% of U.S. participants indicated they would bury/cremate grave goods with a loved one. A comparison between Studies 2 and 3 suggests that the deposition of DLA, clothing, and flowers is bereaved-driven, and that grave goods are largely chosen for their sentimental, emotional, and relationship value, rather than their functional, economic, social, or cultural value. Unlike in Studies 1 and 2, there was no effect of gender, suggesting that the prior effects were deceased-driven. An effect of religiosity revealed that bereaved religious participants are partially motivated to leave grave goods that visibly apply religious values and traditions within social/ritual contexts. As in Study 2, afterlife believers engaged in the practice more than extinctivists did, with Study 3 finding that they are partially motivated by a desire to provision the dead, while intuitive afterlife reasoning motivated both groups equally (91% of the time; the increase from 46% suggests that Study 2 participants may have indicated such beliefs had they elaborated on their reasons). Magical contagion and

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personal comfort/closure were also found to be important motivators, with social/reputational motivations being less important. An association between contagion sensitivity and grave-good deposition was fully explained by a need for ritual completion, suggesting the purpose is to reconstruct the essence of the deceased. Magical contagion was also a reason for not leaving grave goods, but extinctivism was rarely given as a reason here, further suggesting that extinctivist beliefs have little influence on grave-good practices.

Taken together, our studies suggest that intuitive afterlife reasoning—revealed in participants' spontaneous statements or endorsements of prewritten explanations that imply knowledge, feelings, and desires of the decedent (whether the postmortem self or a loved one)—motivates most individuals who participate in grave-good practices, regardless of whether they explicitly deny that consciousness survives death (extinctivists). However, explicit afterlife believers are further motivated by their explicit beliefs to participate in the practice more than extinctivists, with Study 3 suggesting that such beliefs make the need to provision the dead more salient (ironically, given that the vast majority of explicit afterlife believers in Study 3 also thought that the decedent would have everything they needed after death). The results lend credence to the "afterlife use" theory, with the extinctivist findings supporting the notion that humans possess difficult-to-override intuitions about consciousness surviving death (Bering, 2006), a cognitive predisposition that has likely operated since at least the Paleolithic era (i.e., the earliest known grave-good practices). Although the evidence we provide is only indirect, an early emergence for such a predisposition is implied by the evolutionary foundations of human cognition (Heyes, 2012) and by the archaeological record, which contains numerous examples across distinct cultures of grave goods intended for afterlife use (Härke, 2014; Ikram, 2015; Kellehear, 2007; Paulsen, 1967).

Our findings suggest that other important reasons for grave-good practices are secondary or related to this purpose. Regarding personal comfort, for example, afterlife believers were found to derive more comfort from their grave-good practices than did extinctivists, and it likely is comforting to envisage the deceased retaining possession of treasured objects. Magical contagion may also bring comfort if it involves reconstructing the person who was lost with physical remnants of their essence (Newman et al., 2011) and, while this essence might not extend to an entire consciousness, it does appear to represent a very basic kind of psychological immortality.

Although other reasons for grave-good practices were found to be less important, some reasons may be less socially desirable, accessible, or rationalizable than others (e.g., social signaling about one's wealth or alignment with religious norms), which could be investigated in future work that is less reliant on self-report measures. For example, experimental designs that include behavioral-dependent measures—such as a task in which participants are given an opportunity to leave items, or not, for a dead character who has been ascribed beliefs, needs, or desires while they were still alive—may complement survey or interview methods. The reasons explored in the present work may also not be exhaustive. For example, although we included some specific ways to honor the deceased in Table 1, a broader, more ambiguous statement (e.g., "to pay my respects") might have garnered more support, although it would have been difficult to distinguish it from (and it would potentially obscure) other motivations that might be less socially desirable, accessible, or rationalizable (i.e., it begs the

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questions "*why* and *to whom* are you paying respects?"). Researchers might consider using semi-structured interviews to address these questions and to supplement the present work by interrogating what participants mean when they say that a grave good is "part of" a person or is "with them" after death, which may help to clarify the extent to which they are deploying magical contagion beliefs and intuitive afterlife reasoning or are merely speaking figuratively. (Our findings with the CSS and ABS scales suggest it is the former, but we acknowledge some remaining ambiguity on this important issue still.)

In general, NZ and U.S. grave goods were similar, regardless of cultural and methodological differences. Our studies suggest that the prevalence of jewelry as a prehistoric and contemporary grave good can be explained by its embodiment of sentimental, emotional, and relational value, and that photographs have become a contemporary grave good for the same reasons. It seems likely that grave goods in future eras will utilize the same value system, perhaps including more electronic devices (e.g., phones and memory drives) and body-integrated technology that could be seen as essential in the afterlife or as containing more of the deceased's essence (Gibson, 2010).

Our cross-cultural conclusions are nevertheless limited by our Western focus. Our results suggest that grave-good deposition should be universal except when social norms or dictums prevent or discourage it. Such norms were present in early Christendom when burials rarely included grave goods (Härke, 2014). Belief in the Christian paradisaical afterlife may also reduce the need for grave goods relative to other cultures. These considerations aside, we would expect explicit afterlife belief within a culture to partially reflect the ubiquity of grave-good practices (beyond that motivated by intuitive afterlife reasoning) and, thus, that grave-good practices are more prevalent outside of the West.

6. Conclusion

With few exceptions, grave goods have been a popular funeral custom throughout history. Our work has found that the practice persists today, regardless of declining rates of stated afterlife belief, and that this can be largely attributed to our species' cognitive bias for (mis)attributing mental states to dead agents (possibly due to an inability to simulate an absence of consciousness, see Bering, 2006; Nichols, 2007) and its role in fostering a desire to furnish dead bodies with grave goods that meet presumed epistemic and emotional needs.

Notes

1 A low response rate was expected. Beyond the opportunity to take part in what we presented as innovative research, we did not provide a financial incentive for the funeral home staff to participate in a necessarily protracted and demanding study (even though we made every effort to minimize the time demands).
2 Service type refers to whether the service was a burial or a cremation.
3 It is possible that funeral homes that receive more grave goods were more likely to participate in our study, exaggerating the true number of funerals that include them.

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However, all participating funeral homes appeared typical and representative of NZ culture in terms of inclusiveness, accessibility, and affordability. An analysis of funeral home variation can be found in the Supplementary Materials.

4 Despite being asked to record the religion of the deceased, some funeral directors would (perhaps understandably) avoid this question and record only the type of service, for example, "a civil service."

5 This procedure ensured that enough extinctivists took part in the study. Extinctivists are less common than nonextinctivists in U.S. samples. Extinctivists identified in prescreening were, therefore, incentivized to take part.

6 No participants were added during the analysis. We recruited more than the required number of participants in the anticipation of exclusions.

7 Based on responses to the Afterlife Beliefs Scale (see below).

8 The hypothesis-naïve research assistant received the basic definition of afterlife reasoning provided in the text. Their coding was 91.0% and 91.5% in agreement with the two coders who were part of the research team. To prevent bias, all coders saw only participants' list of grave goods and their reasons for wanting them. No other results (e.g., afterlife belief) were included on the coding sheet.

9 We could not measure the extent to which participants were motivated by explicit versus implicit afterlife beliefs, given that only six (3%) participants explicitly mentioned an afterlife. Some participants may have been reluctant to share these beliefs with a scientist who might be skeptical of them (e.g., as one participant noted, "I know there is no logical reason to bury a photo with me. But I suppose it is like taking them with me"). Additionally, it may be seen as unnecessary to state explicit afterlife beliefs if they are implied otherwise (e.g., "Not that I'm still in the grave, as I'm not, but I feel it would be nice to have their pictures in the casket with me"). Furthermore, even with the level of detail reluctantly provided by many participants, it would still be difficult to confirm that their answers were driven by explicit afterlife beliefs without further clarification, for example, by interrogating the statements "taking them with me" or not being "in the grave."

10 Participants were informed (accurately) that "many people are buried with things." Our next study did not use normative language and found similar rates of grave good preference, although there were other methodological differences between the studies.

11 No participants were added during the analysis. We recruited more than the required number of participants in the anticipation of exclusions.

12 There was also an "Other reasons not listed here" statement that directed participants to write their reasons in a text box.

13 We sought a deeper understanding of some motivations. For example, "to honor the deceased's wishes" begs the question: why honor their wishes? Thus, the options "so that they will be at rest" (implicit afterlife belief) and "so that the family will be comforted" (other comfort; see below) were added to represent afterlife and extinctivist forms of reasoning equally. These implicit afterlife belief statements were consistent with the Study 2 coding methods (e.g., a consciousness that "will be" in some state, or an object that is "with them").

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14 The full question was "Before or at a funeral, mourners sometimes place objects in the casket with the deceased, or on their person, which are then buried or cremated with them. They do so for a variety of reasons. For the individual whose funeral you are thinking about, are there any objects that you would place in their casket, or on their person, to be buried or cremated with them?"
15 The descriptions in parentheses were shown to participants.
16 Inattention can be a problem for internet studies (Dance, 2015). For example, one participant completed the CSS in 11 s. Another selected their Table 1 statements in 4 s. Although most participants gave the materials sufficient attention, it would reduce data quality to include participants who did not read them. We employed these time constraints on the basis that it would not be possible to read and respond to the materials in less time.
17 Binomial tests comparing both results to a 50% proportion were significant, $ps < .001$.
18 Religious items were less common in Studies 1 and 2, which is why our coding scheme did not include a category for them.
19 No nonreligious participants indicated social signaling. The odds-ratio, therefore, replaces this zero with a one.
20 This result was qualitatively unchanged when agnostics were included in either group ($ps < .05$).
21 After zeroing, the contagion sensitivity scale had a scale range of 0–176.
22 Seven participants selected "other reasons not listed here" and wrote a text response. These were not coded as they overlapped with the Table 1 statements, and six of the participants had selected alternatives from Table 1.
23 The low endorsement for some statements in Table 1 and for the "social signaling" category, which includes five statements, further suggest that statement frequency did not adversely affect the results, and that participants (who passed our exclusion criteria) were not selecting statements randomly.
24 Some of the results summarized here are reported in the Supplementary Materials.

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